FACTSHEET 1 • INTRODUCTION TO DATA ANALYSIS

Know your variable before you touch it

Every analysis decision starts here. Get the variable type wrong and every number after it is wrong.

Before you calculate anything, ask one question about each column in your dataset: **what kind of variable is this?** The answer decides which summary you may use, which chart you may draw, and which test you may run. Nothing else in analysis is this important.

The four types

Type	What it is	Examples from your data	How you summarise it
Nominal (categories, no order)	Named groups. No group is 'higher' than another.	Sex, municipality, TB type, HIV status	Count (n) and percentage (%)
Ordinal (categories, with order)	Ordered groups, but the gaps between them are not equal or not measurable.	Smear grade (Scanty < 1+ < 2+ < 3+), pain score, satisfaction	Count and %, or median. **Never a mean.**
Discrete (counts)	Whole numbers you can count. No half-values.	Number of visits, days to diagnosis, household size	Median (IQR) usually; mean (SD) if symmetric
Continuous (measures)	Measured on a scale; can take any value including decimals.	Age, weight, height, BMI, distance to facility	Mean (SD) if symmetric; median (IQR) if skewed

Three traps that catch experienced researchers

Trap 1 — Numbers that are really categories

A column can be full of numbers and still be nominal. If your database stores sex as **1 = male, 2 = female**, the average sex is 1.4 — a meaningless number. Ask what the number *represents*, not what it looks like.

Trap 2 — Taking the mean of an ordinal variable

Smear grades run Scanty, 1+, 2+, 3+. That is an order, not a measurement. The step from 1+ to 2+ is not the same size as the step from 2+ to 3+. A 'mean smear grade of 1.8' cannot be interpreted. Report **counts and percentages**, or the median grade.

Trap 3 — Grouping a continuous variable too early

Once you convert age into '<15 / 15-44 / 45+' you can never get the detail back, and you have thrown away statistical power. **Analyse it continuous first**, then group it for presentation if grouping helps the reader.

Your 60-second variable audit

Do this for every column, on Day 1 of your analysis, before anything else:

1. Open the column. Is it **words** or **numbers**?
2. If words — do the categories have a natural order? Yes = ordinal. No = nominal.
3. If numbers — is the number a **quantity**, or a **code standing for a category**? A code is nominal, no matter how numeric it looks.
4. If it is a real quantity — can it take decimals? Yes = continuous. Whole numbers only = discrete.

5. Write your answer in your data dictionary. **Now** you may start analysing.

TIP

Do this audit *with your data dictionary open beside you*, and fill in the 'variable type' column as you go. It takes twenty minutes for a typical dataset and saves you from the single most common analysis error in operational research.

Liafuan importante • Key words in Tetum

Data / dataset	**Dadus**
Variable	**Variavel**
Category / group	**Kategoria / grupu**
Number	**Numeru**
Count	**Konta**
Order, sequence	**ordem, sekuensia**
Age	**Idade**
Sex	**Seksu (feto / mane)**